



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT 8

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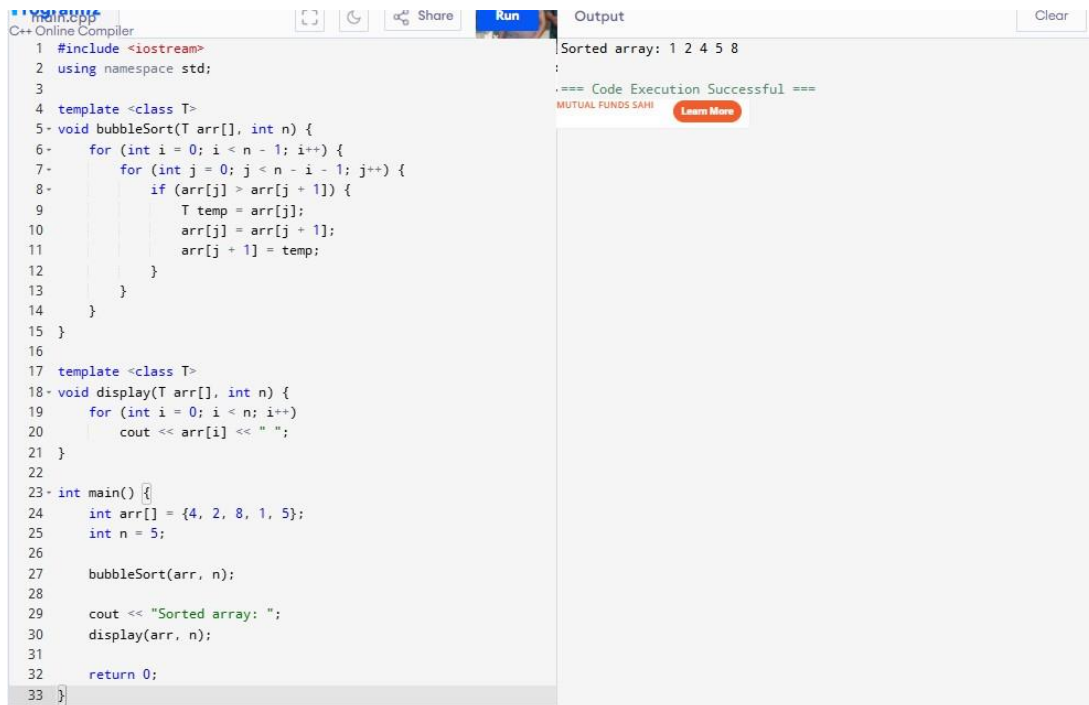
1. Swap two variables of any data type

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 template <class T> 5 void swapValues(T &a, T &b) { 6 T temp = a; 7 a = b; 8 b = temp; 9 } 10 11 int main() { 12 int x = 10, y = 20; 13 14 cout << "Before swap: " << x << " " << y << endl; 15 swapValues(x, y); 16 cout << "After swap: " << x << " " << y << endl; 17 18 return 0; 19 }</pre>	<pre>Before swap: 10 20 After swap: 20 10 === Code Execution Successful ===</pre>

2. Find minimum element in an array

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 template <class T> 5 T findMin(T arr[], int n) { 6 T minVal = arr[0]; 7 8 for (int i = 1; i < n; i++) { 9 if (arr[i] < minVal) 10 minVal = arr[i]; 11 } 12 13 return minVal; 14 } 15 16 int main() { 17 int arr[] = {5, 2, 9, 1, 7}; 18 int n = 5; 19 20 cout << "Minimum element: " << findMin(arr, n); 21 22 return 0; 23 }</pre>	<pre>Minimum element: 1 === Code Execution Successful ===</pre>

3. Sort an array using bubble sort



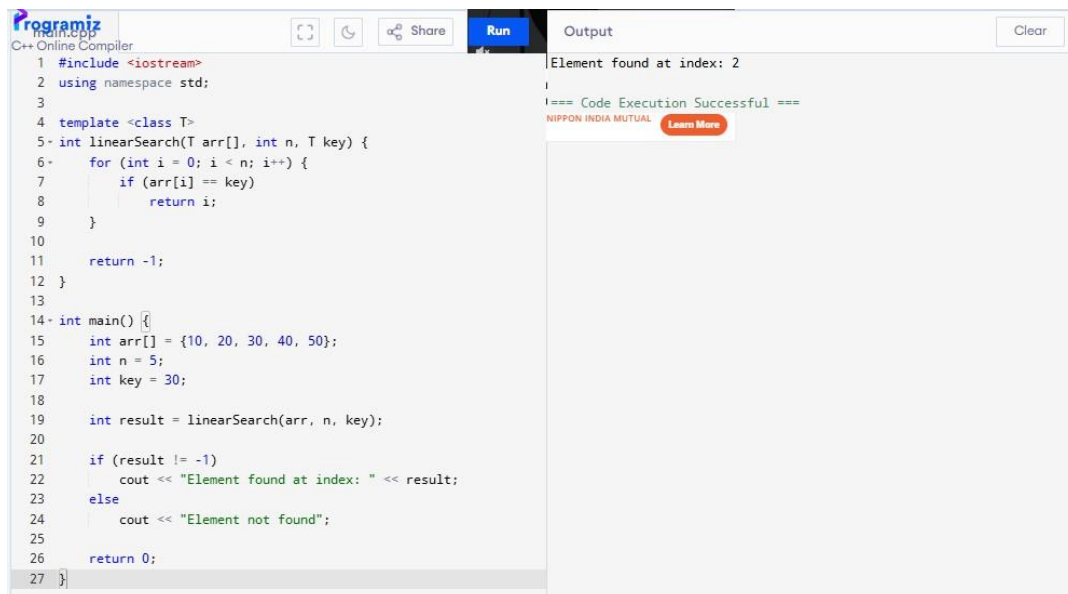
The screenshot shows a C++ program implementing bubble sort. The code is as follows:

```
1 #include <iostream>
2 using namespace std;
3
4 template <class T>
5 void bubbleSort(T arr[], int n) {
6     for (int i = 0; i < n - 1; i++) {
7         for (int j = 0; j < n - i - 1; j++) {
8             if (arr[j] > arr[j + 1]) {
9                 T temp = arr[j];
10                arr[j] = arr[j + 1];
11                arr[j + 1] = temp;
12            }
13        }
14    }
15 }
16
17 template <class T>
18 void display(T arr[], int n) {
19     for (int i = 0; i < n; i++)
20         cout << arr[i] << " ";
21 }
22
23 int main() {
24     int arr[] = {4, 2, 8, 1, 5};
25     int n = 5;
26
27     bubbleSort(arr, n);
28
29     cout << "Sorted array: ";
30     display(arr, n);
31
32     return 0;
33 }
```

The output of the program is:

```
Sorted array: 1 2 4 5 8
==== Code Execution Successful ====
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```

4. Perform linear search



The screenshot shows a C++ program implementing linear search. The code is as follows:

```
1 #include <iostream>
2 using namespace std;
3
4 template <class T>
5 int linearSearch(T arr[], int n, T key) {
6     for (int i = 0; i < n; i++) {
7         if (arr[i] == key)
8             return i;
9     }
10
11     return -1;
12 }
13
14 int main() {
15     int arr[] = {10, 20, 30, 40, 50};
16     int n = 5;
17     int key = 30;
18
19     int result = linearSearch(arr, n, key);
20
21     if (result != -1)
22         cout << "Element found at index: " << result;
23     else
24         cout << "Element not found";
25
26     return 0;
27 }
```

The output of the program is:

```
Element found at index: 2
==== Code Execution Successful ====
NIPPON INDIA MUTUAL Learn More
```

5. Overloaded function template process()

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 template <class T> 5 void process(T a) { 6 cout << "Single parameter: " << a << endl; 7 } 8 9 template <class T> 10 void process(T a, T b) { 11 cout << "Two parameters of same type: " << a << " " << b << endl; 12 } 13 14 template <class T, class U> 15 void process(T a, U b) { 16 cout << "Two parameters of different types: " << a << " " << b << endl; 17 } 18 19 int main() { 20 process(10); 21 process(10, 20); 22 process(10, 5.5); 23 24 return 0; 25 }</pre>	<pre>Single parameter: 10 Two parameters of same type: 10 20 Two parameters of different types: 10 5.5 === Code Execution Successful ===</pre>

Class Templates

1. Stack with push and pop operations

```
main.cpp  [Icons]  Run  Output

1  #include <iostream>
2  using namespace std;
3
4  template <class T>
5  class Stack {
6      T arr[100];
7      int top;
8
9  public:
10     Stack() {
11         top = -1;
12     }
13
14     void push(T value) {
15         if (top == 99)
16             cout << "Stack Overflow" << endl;
17         else
18             arr[++top] = value;
19     }
20
21     void pop() {
22         if (top == -1)
23             cout << "Stack Underflow" << endl;
24         else
25             cout << "Popped element: " << arr[top--] << endl;
26     }
27
28     void display() {
29         if (top == -1)
30             cout << "Stack is empty" << endl;
31         else {
32             for (int i = top; i >= 0; i--)
33                 cout << arr[i] << " ";
34             cout << endl;
35         }
36     }
37 };
38
39 int main() {
40     Stack<int> s;
41
42     s.push(10);
43     s.push(20);
44     s.push(30);
45
46     s.display();
47     s.pop();
48     s.display();
49
50     return 0;
51 }
```

30 20 10
Popped element: 30
20 10

=== Code Execution Successful ===

2. Queue with enqueue and dequeue operations

```
1 #include <iostream>
2 using namespace std;
3
4 template <class T>
5 class Queue {
6     T arr[100];
7     int front, rear;
8
9 public:
10    Queue() {
11        front = -1;
12        rear = -1;
13    }
14
15    void enqueue(T value) {
16        if (rear == 99)
17            cout << "Queue Overflow" << endl;
18        else {
19            if (front == -1)
20                front = 0;
21            arr[++rear] = value;
22        }
23    }
24
25    void dequeue() {
26        if (front == -1 || front > rear)
27            cout << "Queue Underflow" << endl;
28        else
29            cout << "Dequeued element: " << arr[front++] <<
                endl;
30    }
31
32    void display() {
33        if (front == -1 || front > rear)
34            cout << "Queue is empty" << endl;
35        else {
36            for (int i = front; i <= rear; i++)
37                cout << arr[i] << " ";
38            cout << endl;
39        }
40    }
41 };
42
43 int main() {
44     Queue<int> q;
45
46     q.enqueue(10);
47     q.enqueue(20);
48     q.enqueue(30);
49
50     q.display();
51     q.dequeue();
52     q.display();
53
54     return 0;
55 }
```

10 20 30
Dequeued element: 10
20 30

=== Code Execution Successful ===

3. Store and display a pair of values

```
1 #include <iostream>
2 using namespace std;
3
4 template <class T, class U>
5 class Pair {
6     T first;
7     U second;
8
9 public:
10     Pair(T a, U b) {
11         first = a;
12         second = b;
13     }
14
15     void display() {
16         cout << "First value: " << first << endl;
17         cout << "Second value: " << second << endl;
18     }
19 };
20
21 int main() {
22     Pair<int, double> p(10, 5.5);
23
24     p.display();
25
26     return 0;
27 }
```

First value: 10
Second value: 5.5

=== Code Execution Successful ===

4. Basic arithmetic operations

```
1 #include <iostream>
2 using namespace std;
3
4 template <class T>
5 class Arithmetic {
6     T a, b;
7
8 public:
9     Arithmetic(T x, T y) {
10         a = x;
11         b = y;
12     }
13
14     void display() {
15         cout << "Addition: " << a + b << endl;
16         cout << "Subtraction: " << a - b << endl;
17         cout << "Multiplication: " << a * b << endl;
18
19         if (b != 0)
20             cout << "Division: " << a / b << endl;
21         else
22             cout << "Division not possible" << endl;
23     }
24 };
25
26 int main() {
27     Arithmetic<double> obj(10.5, 2.5);
28
29     obj.display();
30
31     return 0;
32 }
```

First value: 10
Second value: 5.5

=== Code Execution Successful ===

5. Input and display array elements

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 template <class T> 5 class Array { 6 T arr[100]; 7 int n; 8 9 public: 10 void input() { 11 cout << "Enter number of elements: "; 12 cin >> n; 13 14 cout << "Enter elements: "; 15 for (int i = 0; i < n; i++) 16 cin >> arr[i]; 17 } 18 19 void display() { 20 cout << "Array elements are: "; 21 for (int i = 0; i < n; i++) 22 cout << arr[i] << " "; 23 } 24 }; 25 26 int main() { 27 Array<int> a; 28 29 a.input(); 30 a.display(); 31 32 return 0; 33 }</pre>	<pre>Enter number of elements: 5 Enter elements: 1 2 3 4 5 Array elements are: 1 2 3 4 5 === Code Execution Successful ===</pre>